Name: Hasnain Ali

Task: 5T OTA Design and Verification on SKY130

5T OTA Design and Verification on SKY130

1. Design Specifications

The objective of this work is to design and verify a single-ended five-transistor Operational Transconductance Amplifier (5T OTA) using the SKY130 CMOS process. The design operates with a 1.8 V supply and drives a 1 pF capacitive load. The amplifier must achieve a minimum DC open-loop gain of 40 dB, a unity-gain bandwidth of 10 MHz and a phase margin of at least 60°. The total supply current must not exceed 20 µA, and the input common-mode range must include 0.9 V.

1.1 Design constraints

Table 1.1: Design specifications.

Parameter	**Target**
Supply voltage	1.8 V
Load capacitance	1 pF
DC open-loop gain	≥ 40 dB
Unity-gain bandwidth	≥ 10 MHz
Phase margin	≥ 60°
Total supply current	≤ 20 µA
Input common-mode voltage	0.9 V must be supported
Maximum supply power	36 µW (20 µA × 1.8 V)
Topology	Single-ended 5-transistor OTA

All design assumptions, transistor dimensions and bias conditions are evaluated using SKY130 device models and circuit simulations.

2. 5T OTA Structure and Transistor Connections

2.1 Topology

The selected architecture is a conventional single-ended 5T OTA. It consists of an NMOS differential input pair, a PMOS current-mirror active load and an NMOS tail-current source. The circuit contains five MOSFETs:

- **M1 and M2 (NMOS differential input pair):** their gates receive the non-inverting and inverting input voltages, respectively. Their sources are connected together to form the common tail node.
- **M3 and M4 (PMOS current-mirror active load):** both sources are connected to the positive supply. M3 is diode-connected, and its gate is connected to the gate of M4. The current mirror provides an active load and converts the differential branch-current variation into a single-ended output.
- **M5 (NMOS tail-current source):** its drain is connected to the common source node of M1 and M2, its source is connected to ground, and its gate receives an external bias voltage V_{BIAS}.

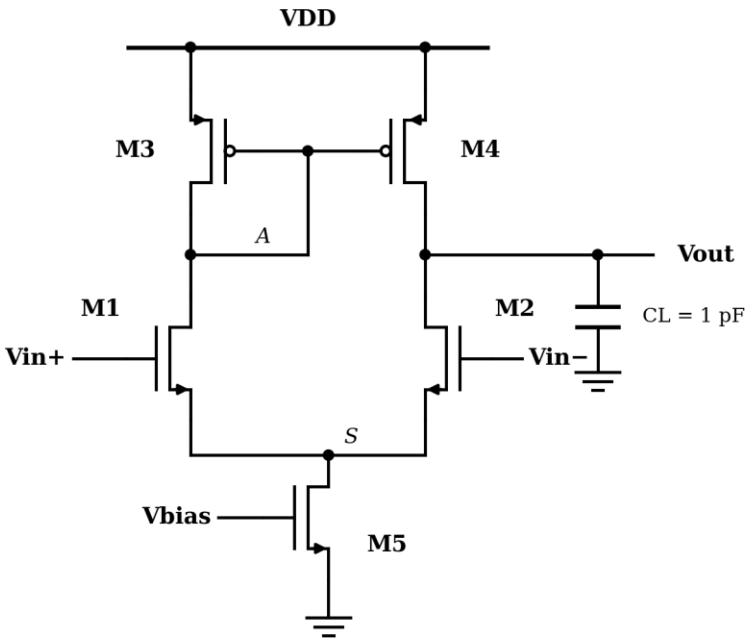


Figure 2.1: Schematic of the single-ended 5T OTA (bulk of NMOS devices tied to GND, bulk of PMOS devices tied to VDD).

2.2 Transistor connections

Table 2.1: Terminal connections and function of each transistor.

Device	Type	Gate	Drain	Source	Function
M1	NMOS	Vin+	Node A	Node S	Input pair (non-inverting side)
M2	NMOS	Vin−	Vout	Node S	Input pair (inverting side)
M3	PMOS	Node A	Node A	VDD	Diode-connected mirror reference
M4	PMOS	Node A	Vout	VDD	Mirror output, active load
M5	NMOS	Vbias	Node S	GND	Tail current source

The output voltage is taken from the common drain connection of M2 and M4. The PMOS bulk terminals are connected to V_{DD}, while the NMOS bulk terminals are connected to ground, following the standard bulk connections for this topology. An NMOS input pair is selected because, for the same W/L and bias current, NMOS devices provide higher transconductance than PMOS devices due to their higher carrier mobility.

2.3 Operating principle

The differential input pair responds to the voltage difference between V_{in+} and V_{in-}. The PMOS current mirror acts as an active load, while M5 establishes the tail current. Variations in the differential input voltage produce changes in the branch currents, resulting in an amplified single-ended output voltage. The transistor-level schematic is implemented and verified using Xschem with the SKY130 PDK.

3. DC Operating Point

3.1 Hand analysis

The DC operating point is analysed with both inputs at the common-mode voltage ($V_{in+} = V_{in-} = V_{CM}$). The input pair and the current mirror are matched, so the circuit is symmetric and the currents divide equally. M3 is diode-connected, which sets the voltage of node A, and the tail node voltage follows from V_{CM} and the gate-source voltage of M1. The saturation conditions of M1 to M5 define the input common-mode range (Figure 3.1).

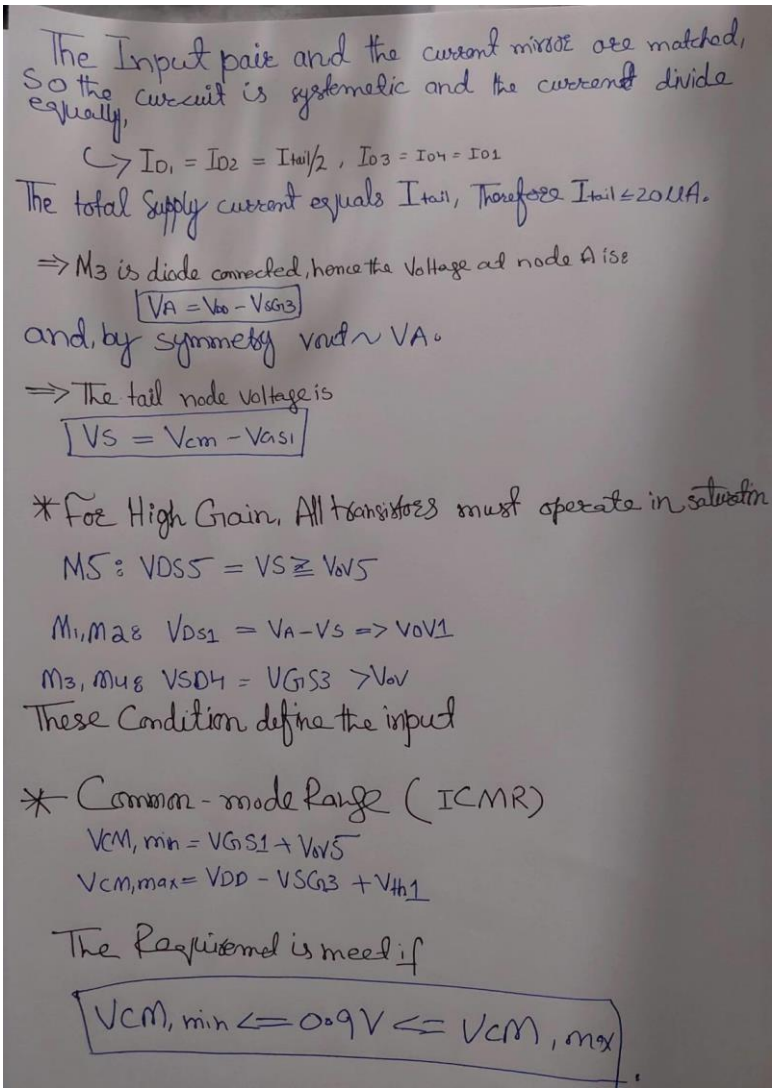


Figure 3.1: Hand analysis of the branch currents, node voltages, saturation conditions and ICMR limits.

4. GBW Requirement and Required Transconductance

4.1 Minimum transconductance

For a single-stage OTA with load capacitance C_L, GBW = g_{m1} / (2π C_L). The minimum transconductance is therefore $g_{m1,min} = 2\pi \times GBW \times C_L = 2\pi \times 10\ MHz \times 1\ pF = 62.83\ \mu S$ (Figure 4.1).

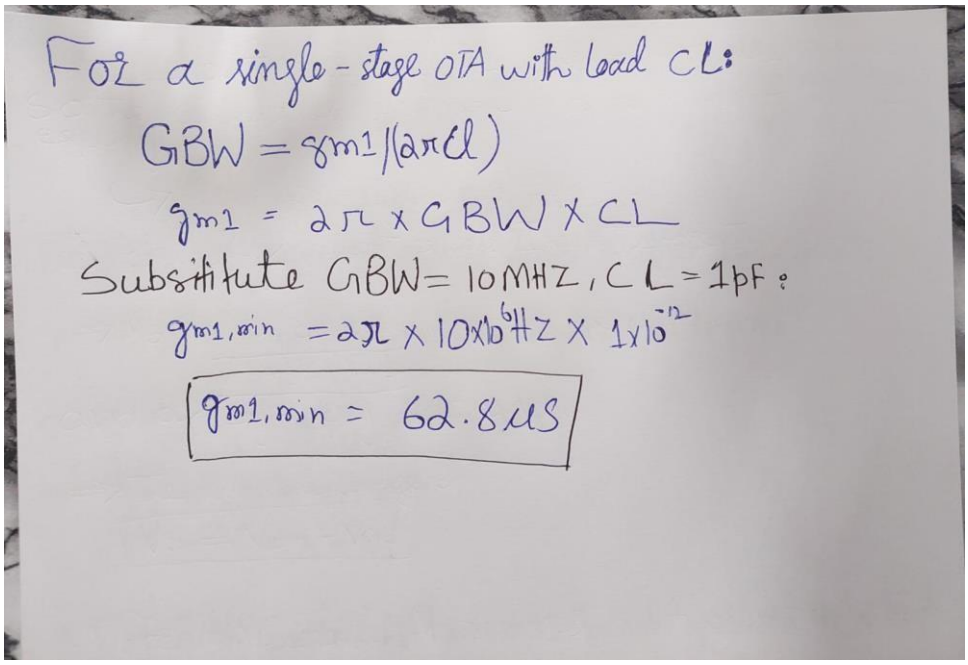


Figure 4.1: Hand calculation of the minimum transconductance.

4.2 Value used in the design

To leave margin for parasitic capacitance at the output node and for model deviations from the hand calculation, g_{m1} = 75 µS (19% above the minimum) is used in the design with a 15 µA tail current. This gives GBW = 75 µS / (2π × 1 pF) ≈ 11.9 MHz.

5. Initial Tail Current Selection

5.1 Initial estimate

With the overdrive voltage assumed as V_{OV} = 0.2 V, the square-law relation $g_m = 2I_D / V_{OV}$ gives $I_{tail} = g_m \times V_{OV}$ = 62.83 µS × 0.2 V = 12.566 µA, a branch current of 6.283 µA and a supply power of 1.8 V × 12.566 µA = 22.62 µW. Both the current (< 20 µA) and the power (< 36 µW) are within the limits (Figure 5.1).

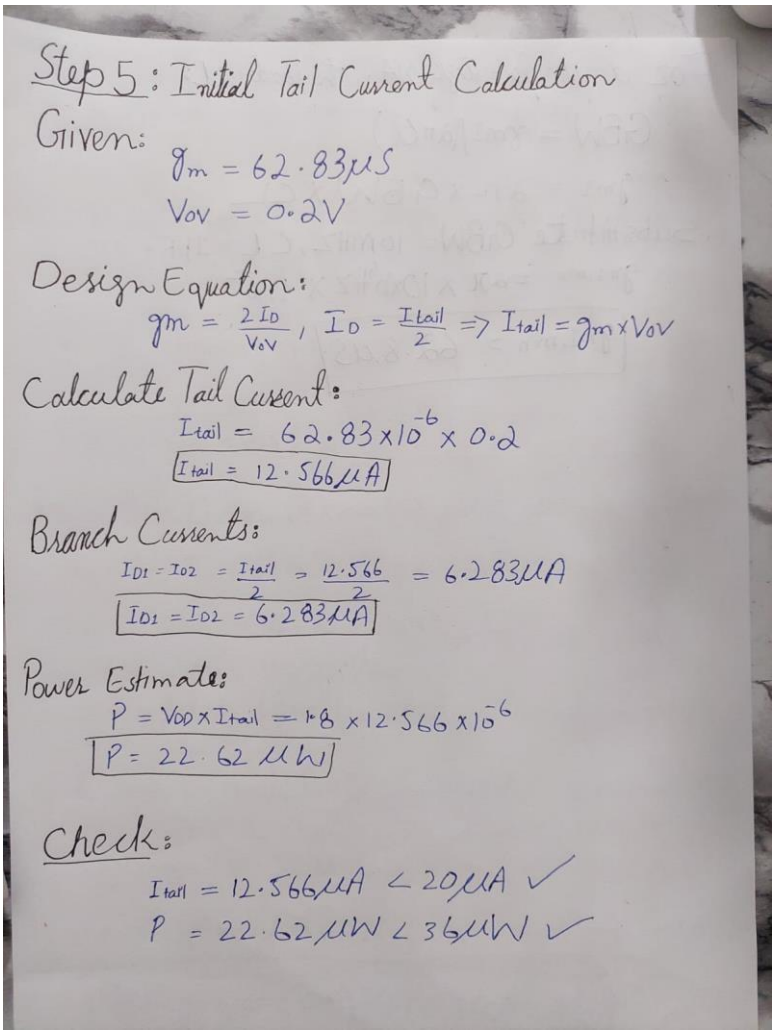


Figure 5.1: Initial tail-current calculation.

5.2 Value carried into the design

The square-law estimate is only a starting point. In Section 7 the design uses g_{m1} = 75 µS and I_{tail} = 15 µA (7.5 µA per branch, g_m/I_D = 10). This keeps the supply current 25% below the 20 µA limit and leaves margin on the GBW. The 15 µA value is used in all later sections.

6. Model Identification

6.1 Objective

Before sizing the OTA, the PDK installation and the device models used by the simulator were identified and verified directly from the terminal. This ensures that every later simulation uses the intended standard 1.8 V devices at a known process corner.

6.2 PDK location

The environment variable PDK_ROOT resolves to /foss/pdks. A search for the sky130A directory returned a path inside the ciel package-manager tree, and a listing of /foss/pdks shows that sky130A is a symbolic link to it. The short path /foss/pdks/sky130A is therefore used throughout. Listing libs.tech confirmed that the ngspice and xschem directories are present (Figure 6.1).

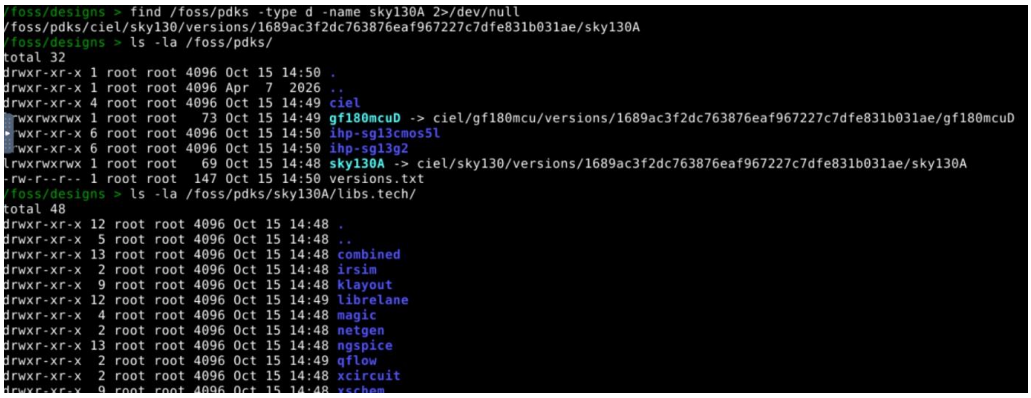


Figure 6.1: PDK root, sky130A directory and libs.tech contents.

6.3 ngspice model folder

The folder libs.tech/ngspice contains the top-level library sky130.lib.spice (16541 bytes) together with the corners and parameters subfolders (Figure 6.2).

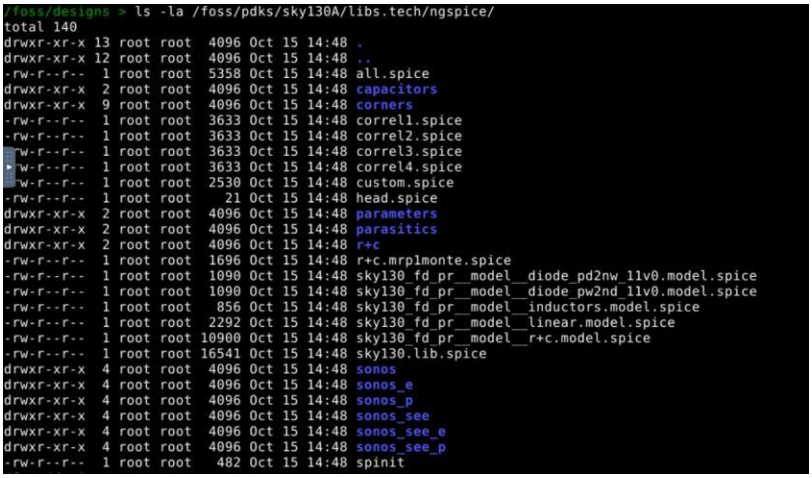


Figure 6.2: Contents of libs.tech/ngspice.

6.4 Process corners

Searching sky130.lib.spice for .lib sections lists the available corners: tt, sf, ff, ss and fs, the combined corners (ll, hh, hl, lh and their sf/ff/ss/fs variants), the mismatch variants (names ending in _mm) and mc (Figures 6.3 and 6.4). The typical corner (tt, line 18) is selected as the nominal condition, since the specifications are defined at nominal process and a 1.8 V supply.

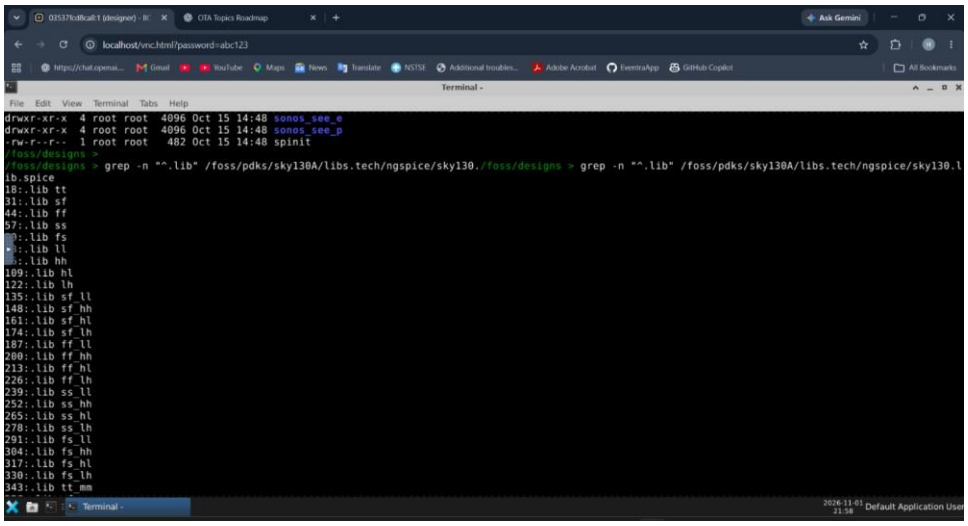


Figure 6.3: .lib sections of sky130.lib.spice (first part).

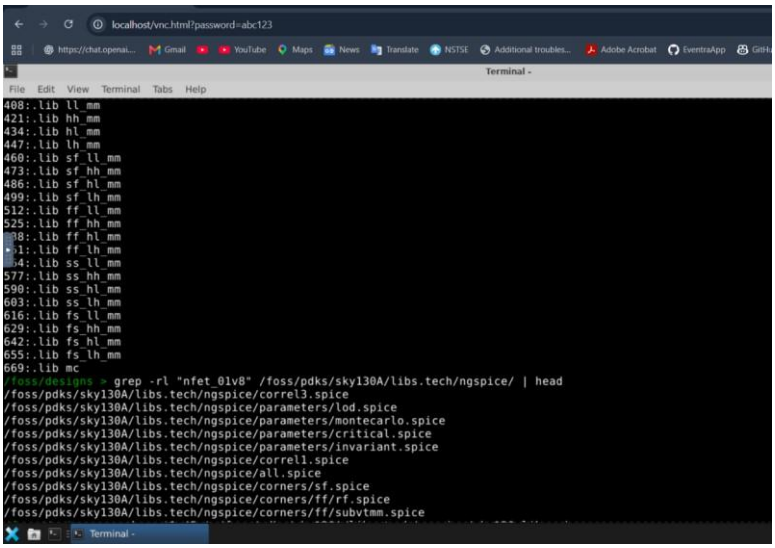


Figure 6.4: Remaining .lib sections, and files referencing nfet_01v8.

6.5 Include structure of the tt corner

Section .lib tt sets mc_mm_switch = 0 and mc_pr_switch = 0 (Monte Carlo disabled) and includes corners/tt.spice for the MOSFET models. As these include paths are relative, the library must always be loaded through sky130.lib.spice with the tt section, not by including corners/tt.spice on its own.

6.6 Device model files

The file corners/tt.spice contains 5 lines referencing nfet_01v8 and 6 referencing pfet_01v8, all of them .include statements pointing to libs.ref/sky130_fd_pr/spice/ (Figure 6.5). The standard 1.8 V models are listed in Table 6.1. The remaining entries (lvt, hvt, ESD and mismatch) are different devices and are not used.

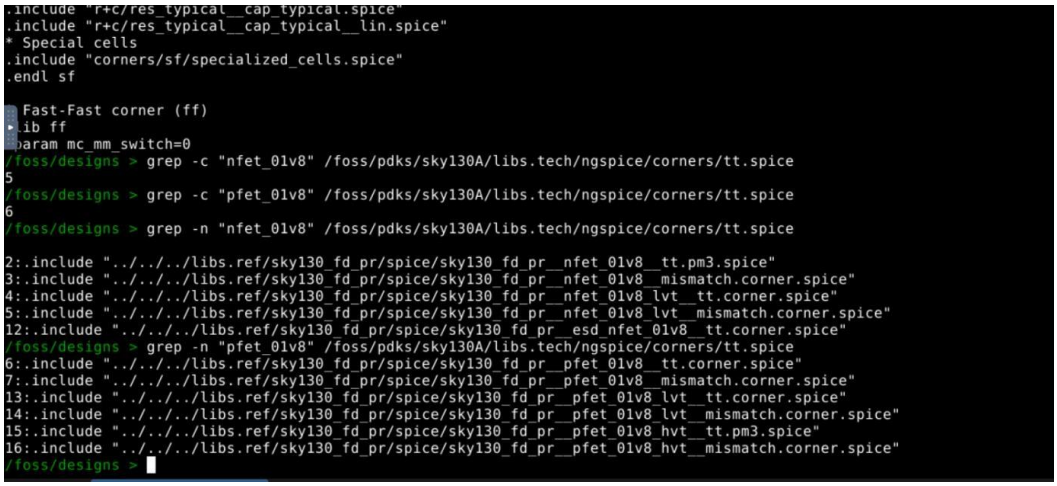


Figure 6.5: nfet_01v8 and pfet_01v8 references in corners/tt.spice.

6.7 Summary

Table 6.1: Verified model files.

Item	Verified result
PDK root / path	PDK_ROOT = /foss/pdks; /foss/pdks/sky130A (symbolic link to the ciel installation)
Model library	libs.tech/ngspice/sky130.lib.spice, section .lib tt (line 18)
Corner / MC switches	Typical (tt); mc_mm_switch = 0, mc_pr_switch = 0 (nominal, no Monte Carlo)
NMOS model	sky130_fd_pr__nfet_01v8 via sky130_fd_pr__nfet_01v8__tt.pm3.spice
PMOS model	sky130_fd_pr__pfet_01v8 via sky130_fd_pr__pfet_01v8__tt.corner.spice
Not used	lvt, hvt, ESD and mismatch variants

6.8 Conclusion

The SKY130A PDK, the typical-corner model library and the standard 1.8 V NMOS (sky130_fd_pr__nfet_01v8) and PMOS (sky130_fd_pr__pfet_01v8) models were located and verified. The NMOS model is used for the input pair (M1, M2) and the tail source (M5), and the PMOS model for the mirror load (M3, M4). The line counts in Figure 6.5 only confirm that the device names are referenced in tt.spice; the chosen W and L values are checked against the model bins by simulation in Section 7.5.

7. Initial Transistor Sizing

7.1 Channel length

The gain requirement $A_v = g_{m1} / (g_{ds2} + g_{ds4}) \geq 100$ (40 dB) limits the total output conductance to $g_{ds2} + g_{ds4} \leq 0.75$ µS for $g_{m1} = 75$ µS. A single NMOS at L = 1 µm already has $g_{ds} \approx 0.47$ µS, which uses 63% of this budget, so L = 2 µm is chosen for M1 to M4 (r_o scales approximately with L). With this length the sum $g_{ds2} + g_{ds4}$ stays close to 0.47 µS, and the PMOS output conductance is assumed equal to the NMOS value (verified later by simulation).

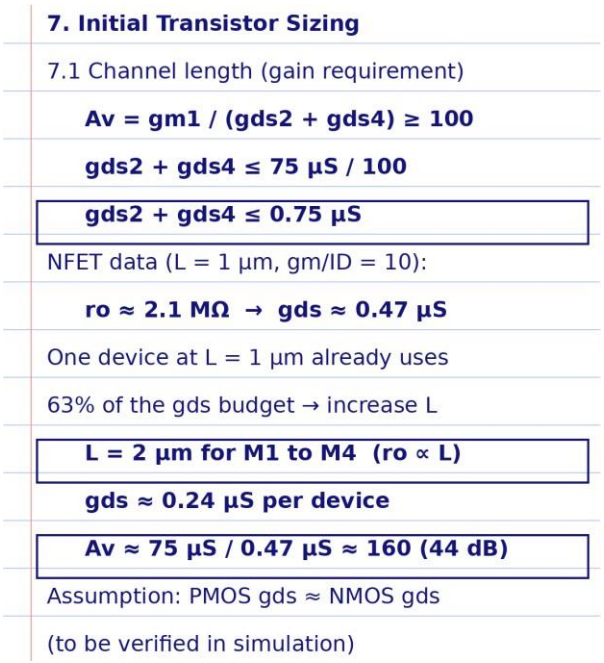


Figure 7.1: Channel-length selection.

The handwritten page rounds the result to $A_v \approx 160$ (44 dB). The unrounded value, 75 µS / 0.476 µS ≈ 157 (43.9 dB), is used in Table 7.2.

7.2 Transistor widths

The input pair is sized from the NFET characterization data at $g_m/I_D = 10$, giving $V_{GS1} \approx 0.775$ V and $W_1 = W_2 = 3.3$ µm. The tail transistor M5 has only $V_S = 0.9\ V - 0.775\ V = 0.125$ V available, so it is sized for low overdrive ($g_m/I_D = 18$), giving $W_5 = 23$ µm at L = 1 µm. The PMOS mirror load is scaled from the NMOS size using $\mu_p \approx 0.35\ \mu_n$, giving $W_3 = W_4 = 9.5$ µm at L = 2 µm. The assumption $\mu_p \approx 0.35\ \mu_n$ is verified later by simulation.

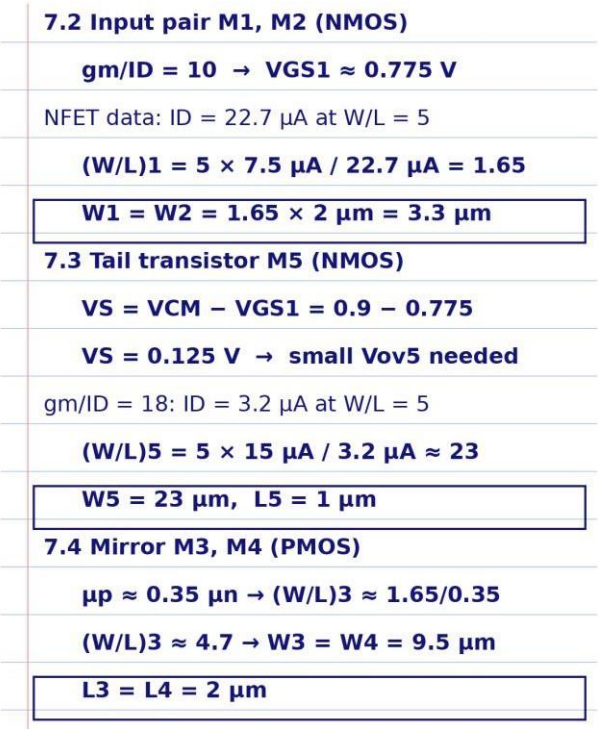


Figure 7.2: Initial W/L sizing.

7.3 Initial dimensions

Table 7.1: Initial transistor dimensions.

Device	Type	W (µm)	L (µm)	Basis
M1, M2	NMOS	3.3	2	g_m/I_D = 10, I_D = 7.5 µA
M3, M4	PMOS	9.5	2	Mirror, $\mu_p \approx 0.35\ \mu_n$
M5	NMOS	23	1	Tail, I_D = 15 µA, low overdrive

7.4 Hand-calculated performance

Table 7.2: Hand-calculated performance estimate.

Parameter	Hand calculation	Target
g_{m1}	75 µS	≥ 62.8 µS
Gain	≈ 157 (43.9 dB)	≥ 40 dB
GBW	≈ 11.9 MHz	≥ 10 MHz
Power	27 µW	≤ 36 µW

The power limit of 36 µW corresponds to the 20 µA current limit at 1.8 V.

7.5 Model bin check

The initial dimensions were checked against the SKY130 model bins by simulating a test circuit (bin_check.cir) with these sizes in ngspice. The simulation completed without model errors, so the chosen W and L values are valid. The printed node voltages come from a rough test bias and are not design results.

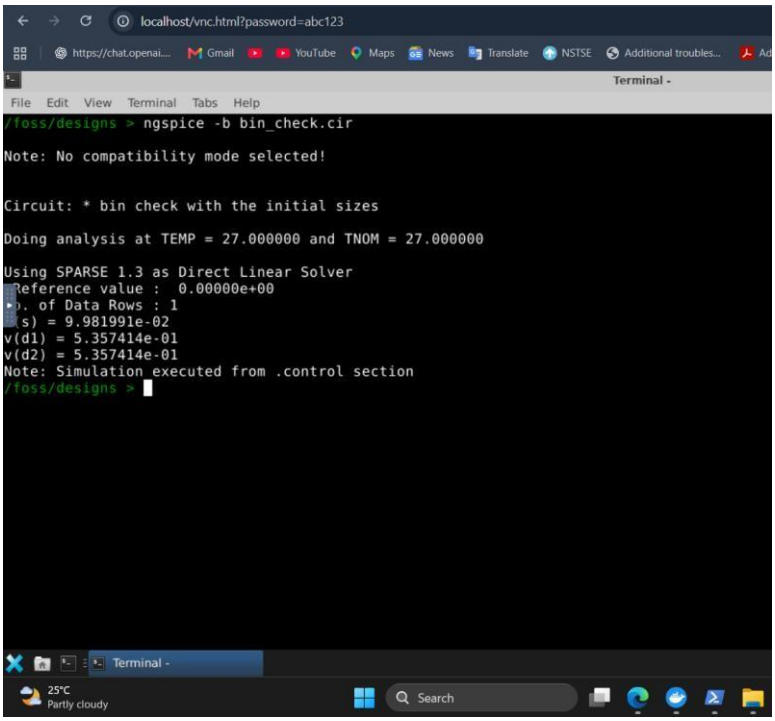


Figure 7.3: Model bin check with the initial transistor sizes.

8. Schematic Entry in Xschem

8.1 Environment setup

The SKY130A environment was activated with the command sak-pdk sky130A. The terminal output confirms PDK_ROOT = /foss/pdks, PDK = sky130A and PDKPATH = /foss/pdks/sky130A, so Xschem and ngspice both load the SKY130 libraries identified in Section 6. Xschem was then started from the same terminal and the SKY130 example schematic top.sch was opened to confirm that the symbol library and the SKY130 menu are available (Figure 8.1).

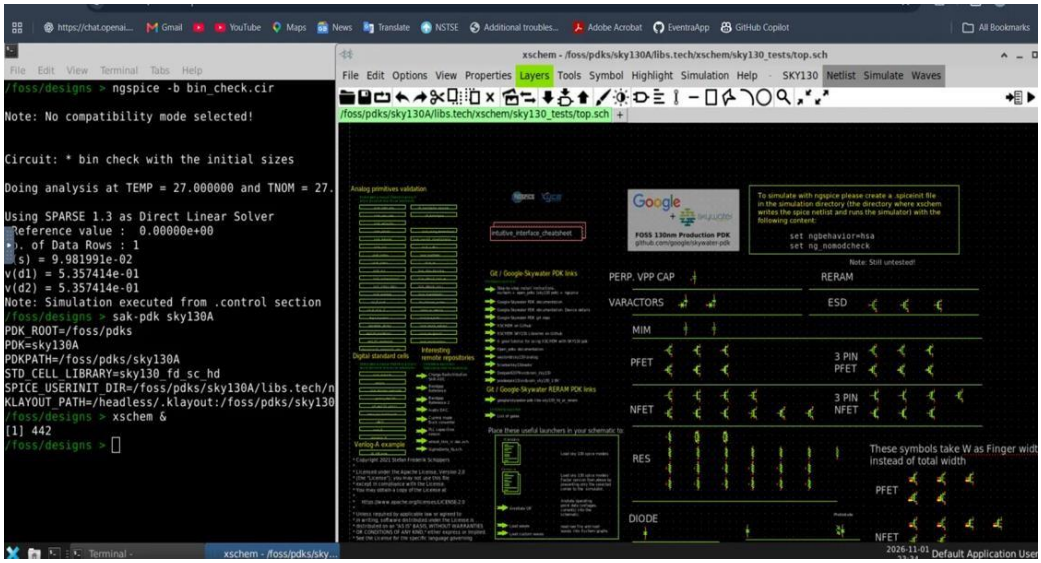


Figure 8.1: sak-pdk sky130A environment and Xschem with the SKY130 example schematic.

8.2 Schematic connections

A new schematic, ota_5t.sch, was created for the 5T OTA. The connections follow the topology of Section 2: an NMOS differential pair (M1, M2) with a PMOS current-mirror load (M3, M4) and an NMOS tail transistor (M5). M3 is diode-connected (gate and drain on node n1) and M4 mirrors its current to the output. The transistor sizes are taken from Table 7.1.

Table 8.1: Device connections and dimensions.

Device	Type	Drain	Gate	Source	Bulk	W / L (µm)
M1	NMOS	n1	inp	tail	GND	3.3 / 2
M2	NMOS	out	inn	tail	GND	3.3 / 2
M3	PMOS	n1	n1	VDD	VDD	9.5 / 2
M4	PMOS	out	n1	VDD	VDD	9.5 / 2
M5	NMOS	tail	vbias	GND	GND	23 / 1

8.3 Placing the devices

Five devices were placed from the SKY130 primitives library: sky130_fd_pr nfet_01v8 for M1, M2 and M5, and sky130_fd_pr__pfet_01v8 for M3 and M4. Width and length were set in the Properties dialog to the values of Table 7.1 (nf = 1 for all devices). NMOS bulk terminals are connected to GND and PMOS bulk terminals to VDD, which keeps all junctions reverse biased.

8.4 Connectivity

Connections were made with net labels and short wires, and nets with the same name are connected by Xschem. M3 is diode-connected (gate and drain on n1) and M4 uses n1 as its gate voltage, which forms the current mirror. M1 and M2 share the node tail, which is the drain of M5. The output node out is the common drain of M2 and M4. The ports are inp, inn and vbias (inputs), out (output), and VDD and GND (supply). Figure 8.2 shows the completed schematic.

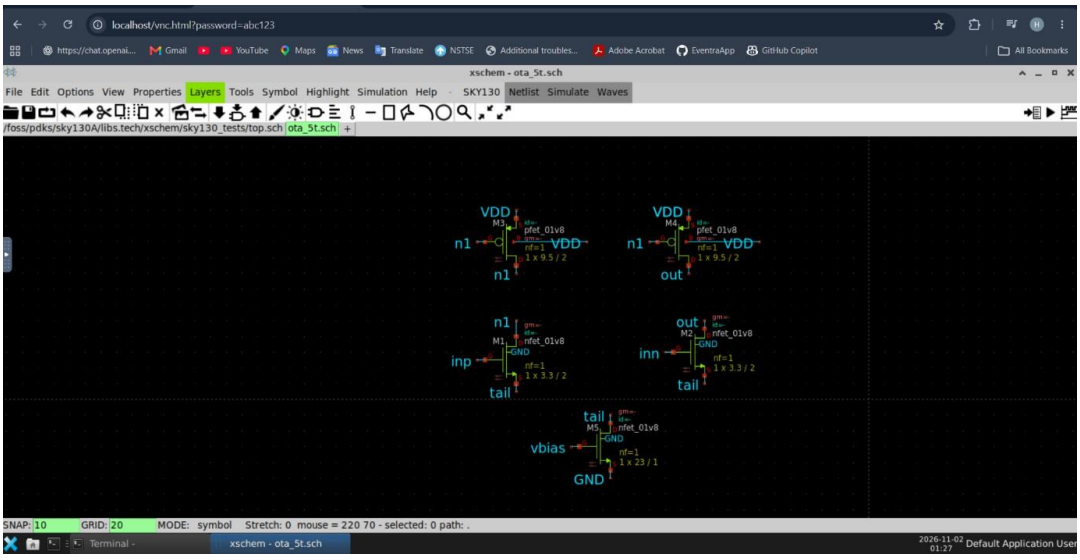


Figure 8.2: Completed 5T OTA schematic in Xschem.

8.5 Netlist verification

The schematic was netlisted with Xschem (Figures 8.3 and 8.4). All five devices match Table 8.1: M1 and M2 share the tail node, M3 is diode-connected, M4 mirrors its current to the output node, and all

dimensions agree with Table 7.1. NMOS bulks are tied to GND and PMOS bulks to VDD. A duplicate net label found during netlisting was removed. Xschem still prints “shorted output node” warnings for the port nets, so the netlist content was checked line by line against Table 8.1.

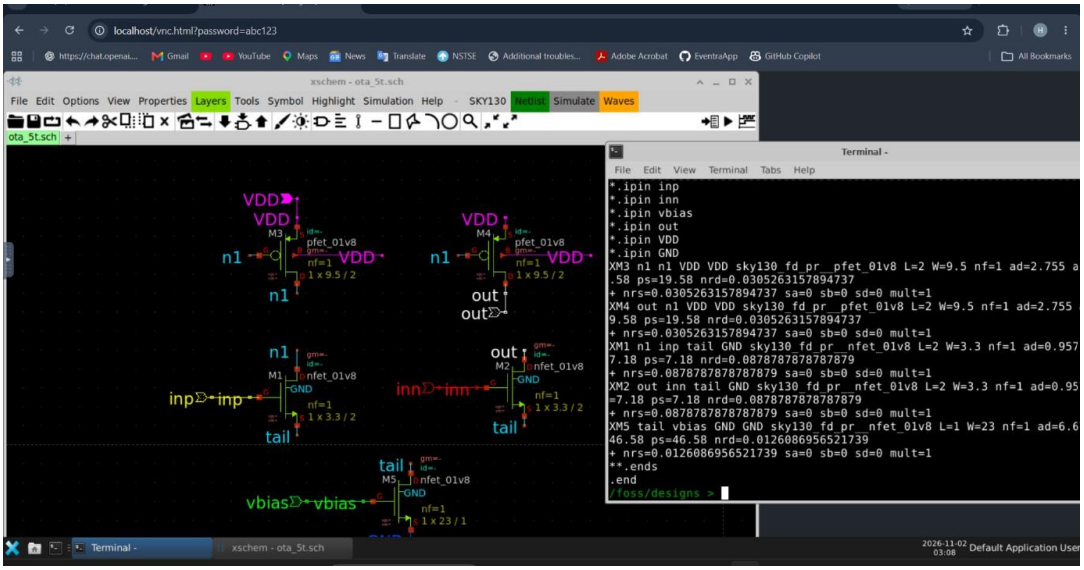


Figure 8.3: Xschem after netlisting the schematic.

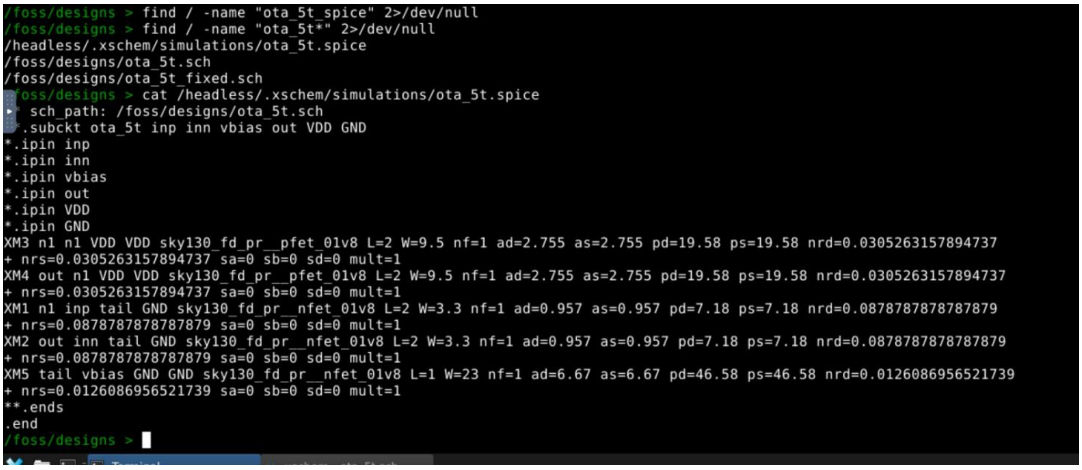


Figure 8.4: Generated netlist ota_5t.spice (XM1 to XM5).

9. Testbench and Simulation Results

9.1 Testbench

The OTA netlist generated from the Xschem schematic was simulated in ngspice with the SKY130 typical-corner models (tt). The OTA is powered from 1.8 V and loaded with C_L = 1 pF at the output. The input common-mode voltage is 0.9 V, and the differential input is applied as ±0.5 V small-signal AC sources on inp and inn, so that the output in dB equals the differential gain. The tail bias voltage vbias is generated by a diode-connected replica of M5 (W/L = 23/1, called M6) fed by a 15 µA ideal current source. This bias branch has its own 1.8 V supply, so the measured supply current is that of the OTA only. The main testbench lines are:

```
.lib /foss/pdks/sky130A/libs.tech/ngspice/sky130.lib.spice tt
.include /foss/designs/devs.txt    (XM1 to XM5 lines from the Xschem netlist)
Vdd VDD 0 1.8
Vb vddb 0 1.8
Vcm cm 0 0.9
```

```
Vinp inp cm dc 0 ac 0.5
Vinn inn cm dc 0 ac -0.5
Ib vddb vbias 15u
XM6 vbias vbias 0 0 sky130_fd_pr__nfet_01v8 L=1 W=23 nf=1 ...
CL out 0 1p
```

```
/foss/designs > grep -n "Vg" /foss/designs/tb_ota.cir
/foss/designs > ngspice /foss/designs/tb_ota.cir
******
** ngspice-47 : Circuit level simulation program
** Compiled with KLU Direct Linear Solver
** The U. C. Berkeley CAD Group
** Copyright 1985-1994, Regents of the University of California.
** Copyright 2001-2026, The ngspice team.
** Please get your ngspice manual from https://ngspice.sourceforge.io/docs.html
** Please file your bug-reports at http://ngspice.sourceforge.net/bugrep.html
** Creation Date: Thu Oct 15 12:58:05 UTC 2026
******

Note: Compatibility modes selected: hs a

Warning: m=xx on .subckt line will override multiplier m hierarchy!


Circuit: * 5t ota testbench, sky130 tt, 1.8 v

option SCALE: Scale is set to 1e-06 for instance and model parameters
Doing analysis at TEMP = 27.000000 and TNOM = 27.000000

Using SPARSE 1.3 as Direct Linear Solver
 Reference value :  0.00000e+00
```

Figure 9.1: ngspice run of the testbench tb_ota.cir.

9.2 DC operating point and AC results

The DC operating point and the AC measurements (gain, unity-gain frequency and phase) are listed in Figure 9.2 and Table 9.1. The tail node voltage of 0.128 V agrees with the hand-calculated V_S = 0.125 V, and the voltages of out and n1 are equal, as expected for a symmetric circuit.

```
/foss/designs > ngspice -b -o /foss/designs/run.log /foss/designs/tb_ota.cir
******
** ngspice-47 : Circuit level simulation program
** Compiled with KLU Direct Linear Solver
** The U. C. Berkeley CAD Group
** Copyright 1985-1994, Regents of the University of California.
** Copyright 2001-2026, The ngspice team.
** Please get your ngspice manual from https://ngspice.sourceforge.io/docs.html
** Please file your bug-reports at http://ngspice.sourceforge.net/bugrep.html
** Creation Date: Thu Oct 15 12:58:05 UTC 2026
******

Batch mode

Comments and warnings go to log-file: /foss/designs/run.log

/foss/designs > grep -E "dcgain|ugf|ph_ugf|^pm|v\(|i\(vdd\)" /foss/designs/run.log
v(out) = 5.753936e-01
v(n1) = 5.753936e-01
v(tail) = 1.284982e-01
v(vbias) = 6.146276e-01
-i(vdd) = 1.361792e-05
dcgain             =  4.05625e+01
ugf                =  1.10957e+07
ph_ugf             =  -9.65407e+01
pm = 8.345934e+01
/foss/designs >
Terminal -    xschem - ota_5t.sch
```

Figure 9.2: DC operating point and AC measurement results from ngspice.

Table 9.1: DC operating point.

Quantity	Simulation
v(out)	0.575 V
v(n1)	0.575 V
v(tail)	0.128 V
v(vbias)	0.615 V
OTA supply current	13.62 µA

The AC analysis from 1 Hz to 1 GHz gives the gain plot in Figure 9.3 and the phase plot in Figure 9.4.

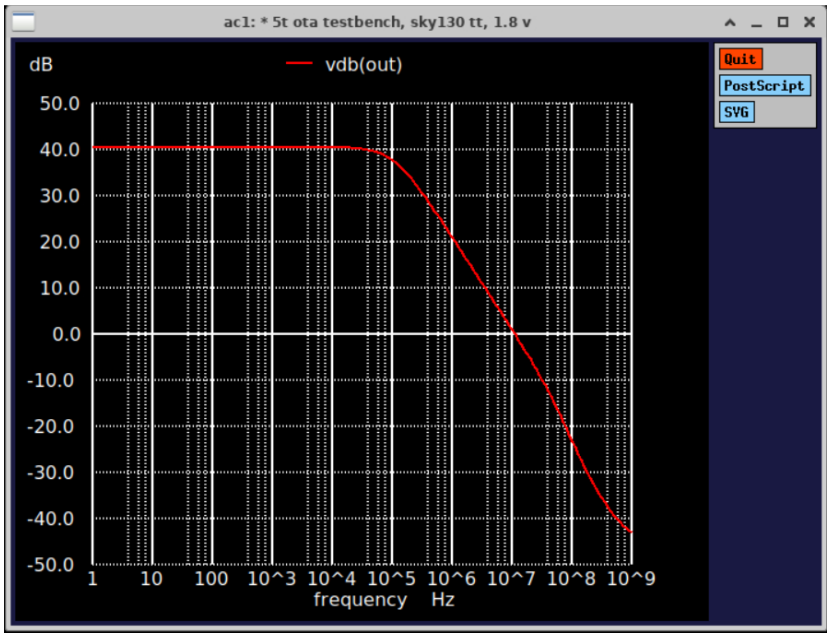


Figure 9.3: Open-loop gain, vdb(out).

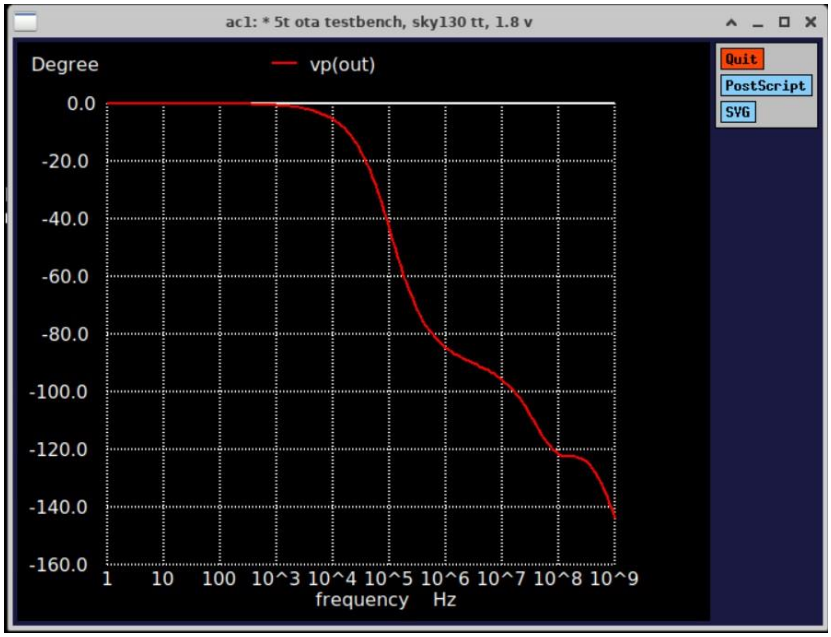


Figure 9.4: Open-loop phase, vp(out).

Table 9.2: Simulation results compared with specifications and hand calculation.

Parameter	**Simulation**	**Specification**	**Hand calculation**
DC gain	40.56 dB	≥ 40 dB	43.9 dB
GBW	11.10 MHz	≥ 10 MHz	11.94 MHz
Phase margin	83.5°	≥ 60°	not calculated
OTA supply current	13.62 µA	≤ 20 µA	15 µA

9.3 Input common-mode range

The input common-mode voltage was swept from 0 V to 1.8 V and the supply current was recorded (Figures 9.5 to 9.7, Table 9.3). The current rises with V_{CM} and saturates at 14.87 µA, which is the value reached when the tail transistor M5 is in saturation. The lower limit of the ICMR is defined as the V_{CM} at which the current falls to 90% of this value (13.38 µA). By linear interpolation between the points at 0.85 V (12.91 µA) and 0.90 V (13.62 µA), this gives a lower limit of about 0.88 V, so 0.9 V is inside the range with a margin of only about 17 mV. At V_{CM} = 0.9 V the current is 13.62 µA, which is 92% of the saturated value, so M5 is operating close to the edge of saturation. The current remains within 3% of its saturated value from 1.2 V up to 1.8 V, so the upper limit was not reached within the sweep. The ICMR was judged from the tail current only; the gain was not re-checked at each V_{CM} point.

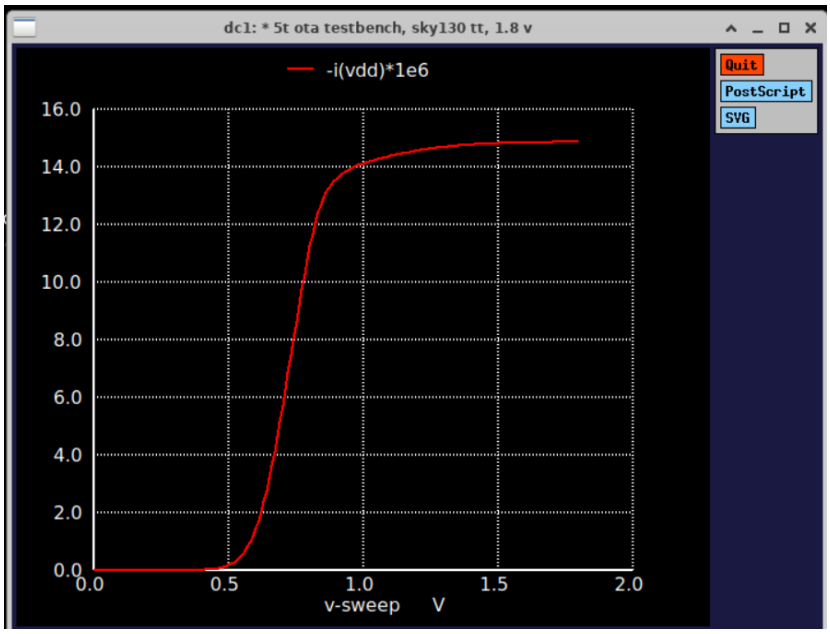


Figure 9.5: Supply current versus input common-mode voltage.

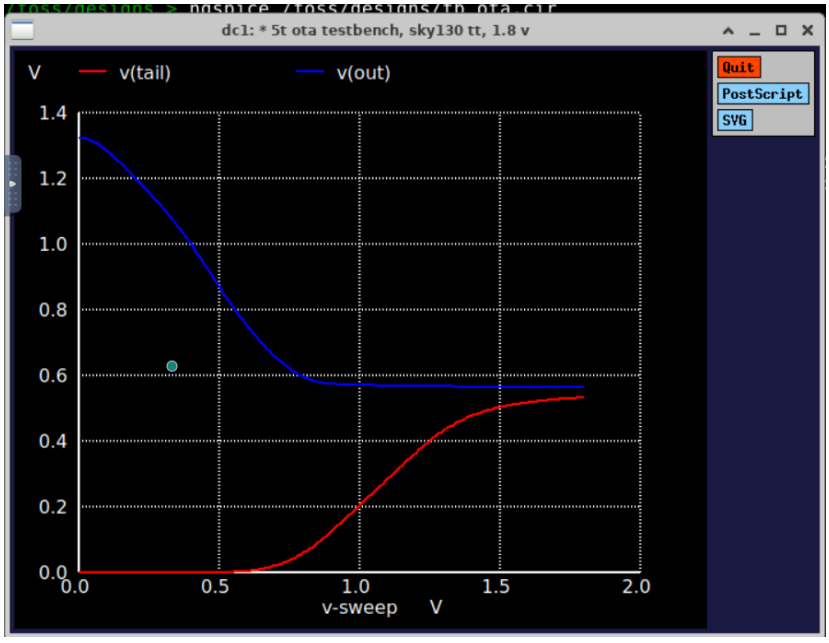


Figure 9.6: v(tail) and v(out) versus input common-mode voltage.

```
/foss/designs > sed '/^\.control/,$d' /foss/designs/tb_ota.cir > /foss/designs/tb_icmr.cir
cat >> /foss/designs/tb_icmr.cir << 'EOF'
.control
dc Vcm 0 1.8 0.01
meas dc i080 find i(Vdd) at=0.80
meas dc i085 find i(Vdd) at=0.85
meas dc i090 find i(Vdd) at=0.90
meas dc i095 find i(Vdd) at=0.95
meas dc i100 find i(Vdd) at=1.00
meas dc i120 find i(Vdd) at=1.20
meas dc i150 find i(Vdd) at=1.50
meas dc i170 find i(Vdd) at=1.70
meas dc i180 find i(Vdd) at=1.80
.endc
.end
EOF
ngspice -b /foss/designs/tb_icmr.cir | grep -E "^i[0-9]"
Warning: m=xx on .subckt line will override multiplier m hierarchy!

Note: Starting dynamic gmin stepping
Note: Dynamic gmin stepping completed
i080                =  -1.12635e-05
i085                =  -1.29106e-05
i090                =  -1.36179e-05
i095                =  -1.39373e-05
i100                =  -1.41289e-05
i120                =  -1.45657e-05
i150                =  -1.48221e-05
i170                =  -1.48611e-05
i180                =  -1.48710e-05
/foss/designs >
```

Figure 9.7: Supply current measured at selected common-mode voltages.

Table 9.3: Supply current versus input common-mode voltage.

V_{CM} (V)	0.80	0.85	0.90	0.95	1.00	1.20	1.50	1.70	1.80
Current (µA)	11.26	12.91	13.62	13.94	14.13	14.57	14.82	14.86	14.87

9.4 Discussion

All four specifications are met at the typical corner, but the DC gain exceeds the 40 dB limit by only 0.56 dB. The simulated gain is 3.3 dB below the hand calculation, which assumed that the PMOS output conductance equals that of the NMOS; this assumption was not verified separately, so the cause of the difference is not established. The supply current of 13.62 µA is lower than the 15 µA design value. The bias replica M6 has V_{DS} = 0.615 V while M5 has V_{DS} = 0.128 V, so channel-length modulation probably makes the mirrored current differ from 15 µA, but this was not isolated by a separate simulation. The lower current also reduces g_m, which is consistent with the GBW being 7% below the hand-calculated value. The bias branch (M6 and the 15 µA source) is not counted as part of the OTA current.

10. Comparison with Specifications and Conclusion

10.1 Comparison

Table 10.1: Final comparison.

Parameter	Specification	Hand calculation	Simulation	Result
DC gain	≥ 40 dB	43.9 dB	40.56 dB	Pass
GBW	≥ 10 MHz	11.94 MHz	11.10 MHz	Pass
Phase margin	≥ 60°	not calculated	83.5°	Pass
Supply current	≤ 20 µA	15 µA	13.62 µA	Pass
ICMR	includes 0.9 V	0.9 V reached	lower limit ≈ 0.88 V	Pass (small margin)

10.2 Conclusion

All specifications are met at the typical corner. The DC gain (0.56 dB margin) and the ICMR lower limit (about 17 mV margin) are close to their limits, and only the typical corner was simulated, so the design may not meet the specifications at other process corners or temperatures.